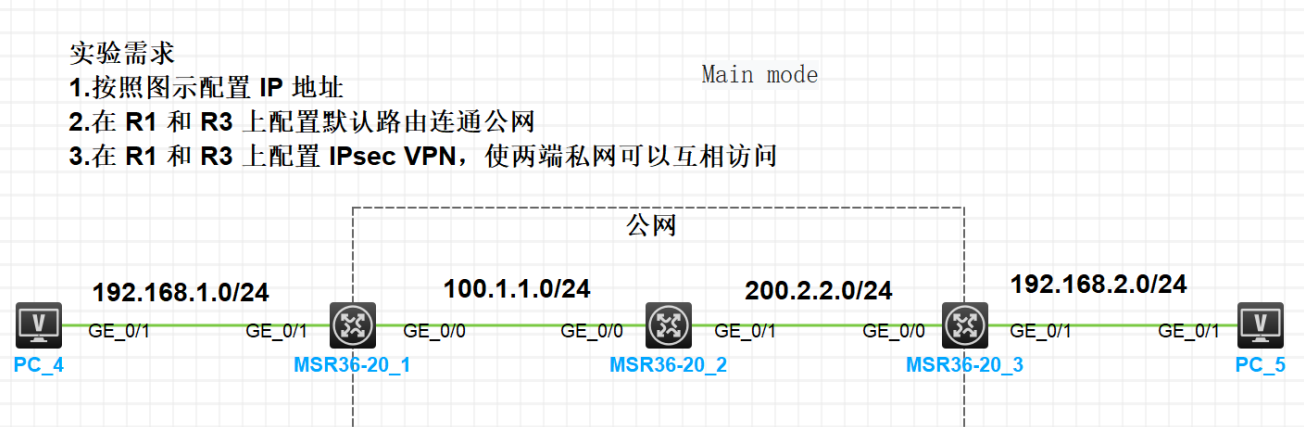


IPsec-VPN-----main_mode

实验拓扑



实验需求

1. 按照图示配置 IP 地址
2. 在 R1 和 R3 上配置默认路由连通公网
3. 在 R1 和 R3 上配置 IPsec VPN，使两端私网可以互相访问

实验配置

```
[R1]int g0/1
[R1-GigabitEthernet0/1]ip ad 192.168.1.1 24
[R1-GigabitEthernet0/1]int g0/0
[R1-GigabitEthernet0/0]ip ad 100.1.1.1 24
[R1]ip route-static 0.0.0.0 0 100.1.1.2

---

[R2]int g0/0
[R2-GigabitEthernet0/0]ip ad 100.1.1.2 24
[R2-GigabitEthernet0/0]int g0/1
[R2-GigabitEthernet0/1]ip ad 200.2.2.2 24
[R2]ip route-static 192.168.1.0 24 100.1.1.1
[R2]ip route-static 192.168.2.0 24 200.2.2.3

---

[R3]int g0/0
[R3-GigabitEthernet0/0]ip ad 200.2.2.3 24
[R3-GigabitEthernet0/0]int g0/1
[R3-GigabitEthernet0/1]ip ad 192.168.2.1 24
[R3]ip route-static 0.0.0.0 0 200.2.2.2
#####
[R1]acl advanced 3000
[R1-acl-ipv4-adv-3000]rule permit ip source 192.168.1.0 0.0.0.255 destination 19
```

```

2.168.2.0 0.0.0.255
[R1]ike proposal 1
[R1-ike-proposal-1]authentication-method pre-share
[R1-ike-proposal-1]encryption-algorithm aes-cbc-128
[R1]ike keychain r3
[R1-ike-keychain-r3]pre-shared-key address 200.2.2.3 key simple 123456
[R1]ike profile r3
[R1-ike-profile-r3]keychain r3
[R1-ike-profile-r3]local-identity address 100.1.1.1
[R1-ike-profile-r3]match remote identity address 200.2.2.3
[R1-ike-profile-r3]proposal 1
[R1]ipsec transform-set r3
[R1-ipsec-transform-set-r3]esp authentication-algorithm sha1
[R1-ipsec-transform-set-r3]esp encryption-algorithm aes-cbc-128
[R1]ipsec policy r3 1 isakmp
[R1-ipsec-policy-isakmp-r3-1]security acl 3000
[R1-ipsec-policy-isakmp-r3-1]ike-profile r3
[R1-ipsec-policy-isakmp-r3-1]transform-set r3
[R1-ipsec-policy-isakmp-r3-1]remote-address 200.2.2.3
[R1-ipsec-policy-isakmp-r3-1]int g0/0
[R1-GigabitEthernet0/0]ipsec apply policy r3

---

[R3]acl advanced 3000
[R3-acl-ipv4-adv-3000]rule permit ip source 192.168.2.0 0.0.0.255 destination 1
92.168.1.0 0.0.0.255
[R3]ike proposal 1
[R3-ike-proposal-1]authentication-method pre-share
[R3-ike-proposal-1]encryption-algorithm aes-cbc-128
[R3]ike keychain r1
[R3-ike-keychain-r1]pre-shared-key address 100.1.1.1 key simple 123456
[R3]ike profile r1
[R3-ike-profile-r1]keychain r1
[R3-ike-profile-r1]proposal 1
[R3-ike-profile-r1]local-identity address 200.2.2.3
[R3-ike-profile-r1]match remote identity address 100.1.1.1
[R3]ipsec transform-set r1
[R3-ipsec-transform-set-r1]esp authentication-algorithm sha1
[R3-ipsec-transform-set-r1]esp encryption-algorithm aes-cbc-128
[R3]ipsec policy r1 1 isakmp
[R3-ipsec-policy-isakmp-r1-1]security acl 3000
[R3-ipsec-policy-isakmp-r1-1]ike-profile r1
[R3-ipsec-policy-isakmp-r1-1]transform-set r1
[R3-ipsec-policy-isakmp-r1-1]remote-address 100.1.1.1
[R3-ipsec-policy-isakmp-r1-1]int g0/0
[R3-GigabitEthernet0/0]ipsec apply policy r1

```

结果验证

连通性测试

```
<H3C>ping 192.168.2.10
Ping 192.168.2.10 (192.168.2.10): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.10: icmp_seq=0 ttl=253 time=1.995 ms
56 bytes from 192.168.2.10: icmp_seq=1 ttl=253 time=1.443 ms
---
<H3C>ping 192.168.1.10
Ping 192.168.1.10 (192.168.1.10): 56 data bytes, press CTRL_C to break
Request time out
56 bytes from 192.168.1.10: icmp_seq=1 ttl=253 time=1.949 ms
56 bytes from 192.168.1.10: icmp_seq=2 ttl=253 time=1.565 ms
```

查看IKE信息

```
<R1>display ike global-info
IKE identity type: --
IKE identity: --
IKE dpd: on-demand
IKE dpd interval (seconds): 0
IKE dpd retry (seconds): 5
IKE invalid-spi-recovery: Disabled
IKE limit max-negotiating-sa: 0
IKE limit max-sa: 0
IKE nat-keepalive (seconds): 20
IKE signature-identity from-certificate: Disabled
IKE keepalive timeout (seconds): 0
IKE keepalive interval (seconds): 0
<R1>display ike proposal
Priority Authentication Authentication Encryption Diffie-Hellman Duration
          method      algorithm    algorithm      group      (seconds)
-----
1          PRE-SHARED-KEY    SHA1          AES-CBC-128   Group 1     86400
default    PRE-SHARED-KEY    SHA1          DES-CBC       Group 1     86400
<R1>display ike statistics
IKE statistics:
No matching proposal: 0
Invalid ID information: 0
Unavailable certificate: 0
Unsupported DOI: 0
Unsupported situation: 0
Invalid proposal syntax: 0
Invalid SPI: 0
Invalid protocol ID: 0
Invalid certificate: 0
Authentication failure: 0
Invalid flags: 0
Invalid message id: 0
Invalid cookie: 0
Invalid transform ID: 0
Malformed payload: 0
```

```

Invalid key information: 0
Invalid hash information: 0
Unsupported attribute: 0
Unsupported certificate type: 0
Invalid certificate authority: 0
Invalid signature: 0
Unsupported exchange type: 0
No available SA: 0
Retransmit timeout: 0
Not enough memory: 0
Enqueue fails: 0
Failures to send R_U_THERE DPD packets: 0
Failures to receive R_U_THERE DPD packets: 0
Failures to send ACK DPD packets: 0
Failures to receive ACK DPD packets: 0
Sent P1 SA lifetime change packets: 0
Received P1 SA lifetime change packets: total=0, process failures=0 (no SA=0,failures to reset
SA soft lifetime=0,failures to reset SA hard lifetime=0)
Sent P2 SA lifetime change packets: 0
Received P2 SA lifetime change packets: total=0, process failures=0
<R1>display ike sa
    Connection-ID    Local                Remote                Flag        DOI
-----
    1                100.1.1.1           200.2.2.3           RD          IPsec
Flags:
RD--READY RL--REPLACED FD-FADING RK-REKEY

-----
<R2>display ike global-info
IKE identity type: --
IKE identity: --
IKE dpd: on-demand
IKE dpd interval (seconds): 0
IKE dpd retry (seconds): 5
IKE invalid-spi-recovery: Disabled
IKE limit max-negotiating-sa: 0
IKE limit max-sa: 0
IKE nat-keepalive (seconds): 20
IKE signature-identity from-certificate: Disabled
IKE keepalive timeout (seconds): 0
IKE keepalive interval (seconds): 0
<R2>display ike pr
<R2>display ike proposal
    Priority Authentication Authentication Encryption Diffie-Hellman Duration
           method      algorithm      algorithm      group      (seconds)
-----
    default PRE-SHARED-KEY    SHA1          DES-CBC        Group 1    86400
<R2>display ike st
<R2>display ike statistics
IKE statistics:
    No matching proposal: 0
    Invalid ID information: 0

    Unavailable certificate: 0

```

```

Unsupported DOI: 0
Unsupported situation: 0
Invalid proposal syntax: 0
Invalid SPI: 0
Invalid protocol ID: 0
Invalid certificate: 0
Authentication failure: 0
Invalid flags: 0
Invalid message id: 0
Invalid cookie: 0
Invalid transform ID: 0
Malformed payload: 0
Invalid key information: 0
Invalid hash information: 0
Unsupported attribute: 0
Unsupported certificate type: 0
Invalid certificate authority: 0
Invalid signature: 0
Unsupported exchange type: 0
No available SA: 0
Retransmit timeout: 0
Not enough memory: 0
Enqueue fails: 0
Failures to send R_U_THERE DPD packets: 0
Failures to receive R_U_THERE DPD packets: 0
Failures to send ACK DPD packets: 0
Failures to receive ACK DPD packets: 0
Sent P1 SA lifetime change packets: 0
Received P1 SA lifetime change packets: total=0, process failures=0 (no SA=0,failures to reset
SA soft lifetime=0,failures to reset SA hard lifetime=0)
Sent P2 SA lifetime change packets: 0
Received P2 SA lifetime change packets: total=0, process failures=0
<R2>display ike sa
      Connection-ID    Local                Remote                Flag        DOI
-----
<R2>

-----
<R3>display ike global-info
IKE identity type: --
IKE identity: --
IKE dpd: on-demand
IKE dpd interval (seconds): 0
IKE dpd retry (seconds): 5
IKE invalid-spi-recovery: Disabled
IKE limit max-negotiating-sa: 0
IKE limit max-sa: 0
IKE nat-keepalive (seconds): 20
IKE signature-identity from-certificate: Disabled
IKE keepalive timeout (seconds): 0
IKE keepalive interval (seconds): 0
<R3>display ike pro

<R3>display ike proposal

```

Priority	Authentication method	Authentication algorithm	Encryption algorithm	Diffie-Hellman group	Duration (seconds)
1	PRE-SHARED-KEY	SHA1	AES-CBC-128	Group 1	86400
default	PRE-SHARED-KEY	SHA1	DES-CBC	Group 1	86400

```

<R3>display ike st
<R3>display ike statistics
IKE statistics:
  No matching proposal: 0
  Invalid ID information: 0
  Unavailable certificate: 0
  Unsupported DOI: 0
  Unsupported situation: 0
  Invalid proposal syntax: 0
  Invalid SPI: 0
  Invalid protocol ID: 0
  Invalid certificate: 0
  Authentication failure: 0
  Invalid flags: 0
  Invalid message id: 0
  Invalid cookie: 0
  Invalid transform ID: 0
  Malformed payload: 0
  Invalid key information: 0
  Invalid hash information: 0
  Unsupported attribute: 0
  Unsupported certificate type: 0
  Invalid certificate authority: 0
  Invalid signature: 0
  Unsupported exchange type: 0
  No available SA: 0
  Retransmit timeout: 0
  Not enough memory: 0
  Enqueue fails: 0
  Failures to send R_U_THERE DPD packets: 0
  Failures to receive R_U_THERE DPD packets: 0
  Failures to send ACK DPD packets: 0
  Failures to receive ACK DPD packets: 0
  Sent P1 SA lifetime change packets: 0
  Received P1 SA lifetime change packets: total=0, process failures=0 (no SA=0,failures to reset
SA soft lifetime=0,failures to reset SA hard lifetime=0)
  Sent P2 SA lifetime change packets: 0
  Received P2 SA lifetime change packets: total=0, process failures=0
<R3>display ike sa
  Connection-ID    Local              Remote              Flag    DOI
-----
  1                200.2.2.3         100.1.1.1          RD      IPsec
Flags:
RD--READY RL--REPLACED FD-FADING RK-REKEY
<R3>

```

查看IPsec信息

```
<R1>display ipsec policy
```

```
-----
```

```
IPsec Policy: r3
```

```
Interface: GigabitEthernet0/0
```

```
-----
```

```
-----
```

```
Sequence number: 1
```

```
Mode: ISAKMP
```

```
-----
```

```
Traffic Flow Confidentiality: Disabled
```

```
Security data flow: 3000
```

```
Selector mode: standard
```

```
Local address:
```

```
Remote address: 200.2.2.3
```

```
Remote address switchback mode: Disabled
```

```
Transform set: r3
```

```
IKE profile: r3
```

```
IKEv2 profile:
```

```
smart-link policy:
```

```
SA trigger mode: Traffic-based
```

```
SA duration(time based): 3600 seconds
```

```
SA duration(traffic based): 1843200 kilobytes
```

```
SA soft-duration buffer(time based): --
```

```
SA soft-duration buffer(traffic based): --
```

```
SA idle time: --
```

```
SA df-bit:
```

```
<R1>display ipsec pro
```

```
<R1>display ipsec profile
```

```
<R1>tr
```

```
<R1>display ipsec tr
```

```
<R1>display ipsec transform-set
```

```
IPsec transform set: r3
```

```
State: complete
```

```
Encapsulation mode: tunnel
```

```
ESN: Disabled
```

```
PFS:
```

```
Transform: ESP
```

```
ESP protocol:
```

```
Integrity: SHA1
```

```
Encryption: AES-CBC-128
```

```
<R1>display ipsec st
```

```
<R1>display ipsec statistics
```

```
IPsec packet statistics:
```

```
Received/sent packets: 9/9
```

```
Received/sent bytes: 864/864
```

```
Dropped packets (received/sent): 0/0
```

```
Dropped packets statistics
```

```
No available SA: 0
```

```
Wrong SA: 0
```

```
Invalid length: 0
```

```
Authentication failure: 0
Encapsulation failure: 0
Decapsulation failure: 0
Replayed packets: 0
ACL check failure: 0
MTU check failure: 0
Loopback limit exceeded: 0
Crypto speed limit exceeded: 0
<R1>display ipsec sa
-----

Interface: GigabitEthernet0/0
-----

-----

IPsec policy: r3
Sequence number: 1
Mode: ISAKMP
-----

Tunnel id: 0
Encapsulation mode: tunnel
Perfect Forward Secrecy:
Inside VPN:
Extended Sequence Numbers enable: N
Traffic Flow Confidentiality enable: N
Transmitting entity: Responder
Path MTU: 1428
Tunnel:
    local address: 100.1.1.1
    remote address: 200.2.2.3
Flow:
    sour addr: 192.168.1.0/255.255.255.0 port: 0 protocol: ip
    dest addr: 192.168.2.0/255.255.255.0 port: 0 protocol: ip

[Inbound ESP SAs]
SPI: 3632618354 (0xd8855b72)
Connection ID: 4294967296
Transform set: ESP-ENCRYPT-AES-CBC-128 ESP-AUTH-SHA1
SA duration (kilobytes/sec): 1843200/3600
SA remaining duration (kilobytes/sec): 1843199/3064
Max received sequence-number: 9
Anti-replay check enable: Y
Anti-replay window size: 64
UDP encapsulation used for NAT traversal: N
Status: Active

[Outbound ESP SAs]
SPI: 1084405812 (0x40a2b834)
Connection ID: 4294967297
Transform set: ESP-ENCRYPT-AES-CBC-128 ESP-AUTH-SHA1
SA duration (kilobytes/sec): 1843200/3600
SA remaining duration (kilobytes/sec): 1843199/3064
Max sent sequence-number: 9

UDP encapsulation used for NAT traversal: N
```



```
Status: Active
<R1>display ipsec g
<R1>display ipsec global-info
IPsec anti-reply check: Enabled
IPsec anti-replay window (packets): 64
IPsec decrypt-check: Enabled
IPsec flow-redirect: Disabled
IPsec fragmentation: before-encryption
IPsec global-df-bit: copy
IPsec limit max-tunnel: --
IPsec redundancy: Disabled
IPsec sa global-duration time-based (seconds): 3600
IPsec sa global-duration traffic-based (KBytes): 1843200
IPsec sa idle-time (seconds): --
<R1>
```

```
-----
<R3>display ipsec global-info
IPsec anti-reply check: Enabled
IPsec anti-replay window (packets): 64
IPsec decrypt-check: Enabled
IPsec flow-redirect: Disabled
IPsec fragmentation: before-encryption
IPsec global-df-bit: copy
IPsec limit max-tunnel: --
IPsec redundancy: Disabled
IPsec sa global-duration time-based (seconds): 3600
IPsec sa global-duration traffic-based (KBytes): 1843200
IPsec sa idle-time (seconds): --
<R3>display ipsec p
<R3>display ipsec p2mp
<R3>display ipsec policy
```

```
-----
IPsec Policy: r1
Interface: GigabitEthernet0/0
-----
```

```
-----
Sequence number: 1
Mode: ISAKMP
-----
Traffic Flow Confidentiality: Disabled
Security data flow: 3000
Selector mode: standard
Local address:
Remote address: 100.1.1.1
Remote address switchback mode: Disabled
Transform set: r1
IKE profile: r1
IKEv2 profile:
smart-link policy:
SA trigger mode: Traffic-based

SA duration(time based): 3600 seconds
```

```
SA duration(traffic based): 1843200 kilobytes
SA soft-duration buffer(time based): --
SA soft-duration buffer(traffic based): --
SA idle time: --
SA df-bit:
<R3>display ipsec st
<R3>display ipsec statistics
IPsec packet statistics:
  Received/sent packets: 9/9
  Received/sent bytes: 864/864
  Dropped packets (received/sent): 0/1

Dropped packets statistics
  No available SA: 1
  Wrong SA: 0
  Invalid length: 0
  Authentication failure: 0
  Encapsulation failure: 0
  Decapsulation failure: 0
  Replayed packets: 0
  ACL check failure: 0
  MTU check failure: 0
  Loopback limit exceeded: 0
  Crypto speed limit exceeded: 0
<R3>display ipsec tr
<R3>display ipsec transform-set
IPsec transform set: r1
  State: complete
  Encapsulation mode: tunnel
  ESN: Disabled
  PFS:
  Transform: ESP
  ESP protocol:
    Integrity: SHA1
    Encryption: AES-CBC-128

<R3>display ipsec sa
-----
Interface: GigabitEthernet0/0
-----

-----
IPsec policy: r1
Sequence number: 1
Mode: ISAKMP
-----

Tunnel id: 0
Encapsulation mode: tunnel
Perfect Forward Secrecy:
Inside VPN:
Extended Sequence Numbers enable: N
Traffic Flow Confidentiality enable: N

Transmitting entity: Initiator
```

Path MTU: 1428

Tunnel:

local address: 200.2.2.3

remote address: 100.1.1.1

Flow:

sour addr: 192.168.2.0/255.255.255.0 port: 0 protocol: ip

dest addr: 192.168.1.0/255.255.255.0 port: 0 protocol: ip

[Inbound ESP SAs]

SPI: 1084405812 (0x40a2b834)

Connection ID: 4294967296

Transform set: ESP-ENCRYPT-AES-CBC-128 ESP-AUTH-SHA1

SA duration (kilobytes/sec): 1843200/3600

SA remaining duration (kilobytes/sec): 1843199/2988

Max received sequence-number: 9

Anti-replay check enable: Y

Anti-replay window size: 64

UDP encapsulation used for NAT traversal: N

Status: Active

[Outbound ESP SAs]

SPI: 3632618354 (0xd8855b72)

Connection ID: 4294967297

Transform set: ESP-ENCRYPT-AES-CBC-128 ESP-AUTH-SHA1

SA duration (kilobytes/sec): 1843200/3600

SA remaining duration (kilobytes/sec): 1843199/2988

Max sent sequence-number: 9

UDP encapsulation used for NAT traversal: N

Status: Active

<R3>